



Trends in cocaine and crystal methamphetamine injection over time in a Canadian setting between 2008 and 2018

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ARTICLE INFO

Keywords:

Methamphetamine

Cocaine

Substance-related disorders

Substance use

Intravenous

ABSTRACT

Objective: Unregulated stimulant use is rising globally, though trends in use of cocaine and crystal methamphetamine (CM), the two most commonly used unregulated stimulants in North America, are poorly characterized in many settings. In this study we examined patterns and associations between the injection of cocaine and CM over time in an urban Canadian setting.

Methods: The study collected data from two prospective cohorts of **people who inject drugs in Vancouver, Canada, between 2008 and 2018**. We applied a time series analysis, using multivariable linear regression to detect relationships between reported CM and cocaine injection and year, controlling for covariates. The study used cross-correlation to evaluate the relative trajectories of each substance over time.

Results: **Among 2056 participants, per annum rates of reported injection cocaine use declined significantly over the duration of this study from 45 % to 18 % ($p < 0.001$), while rates of CM injection increased from 17 % to 32 % ($p < 0.001$).** Multivariable linear regression showed that recent CM injection was negatively associated with recent cocaine injection ($\beta = -0.609$, 95 % CI = $-0.750, -0.467$). Cross-correlation demonstrated that injection CM use was associated with a decrease in the likelihood of cocaine injection 12 months later ($p = 0.002$).

Conclusions: These findings demonstrate an epidemiological shift in patterns of injection stimulant use, with rising CM injection associated with a corresponding decrease in cocaine injection over time. Strategies are urgently needed that help to treat and reduce harm among the growing population of people who inject CM.

1. Introduction

Stimulant use is on the rise globally, representing one of the most widely used categories of unregulated drugs around the world, second only to cannabis (United Nations, 2019). Though the US National Survey on Drug Use and Health showed that rates of stimulant use have remained relatively steady over the past decade, amphetamine-related hospitalizations and health care expenditures in the United States have been increasing over that same time span, and the involvement of methamphetamine and other psychostimulants in overdose deaths has risen markedly both in the United States and Canada (British Columbia Coroners Service, 2021; Jones, Bekheet, et al., 2020; Jones, Olsen, et al., 2020; Kariisa et al., 2019; Substance Abuse and Mental Health Services Administration, 2020; Winkelman et al., 2018). Rates of age-adjusted drug overdose deaths involving cocaine decreased from 2006 to 2012

but have been climbing since that time, while seizures of cocaine by the US Drug Enforcement Agency have also increased over the past ten years (Kariisa et al., 2019; U.S. Department of Justice, 2019). Internationally, the burden of unregulated stimulants such as cocaine and amphetamines is felt to be rising in many settings, and the lack of a clear understanding of prevalence, treatment options, and effective policy responses has been highlighted as a global issue (Farrell et al., 2019).

Despite the increasing morbidity and mortality related to stimulant use, we know little about the relationships between patterns and trajectories of injection stimulant use over time, specifically cocaine and crystal methamphetamine (CM). Injection stimulant use is of particular importance due to its association with increased overdose rates, HIV and hepatitis C transmission and disease progression, and other complicated infections such as abscesses and infective endocarditis (Al-Tayyib et al., 2017; Farrell et al., 2019; Lloyd-Smith et al., 2005; Serota et al., 2020).

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<https://doi.org/10.1016/j.josat.2023.208982>

Received 25 January 2022; Received in revised form 1 January 2023; Accepted 12 February 2023

Available online 25 February 2023

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The significant and severe consequences specifically associated with stimulants used by injection route (i.e. infectious disease and overdose risk) make it a unique public health priority (Compton et al., 2020). Further, among populations of people who inject drugs, increasing reports of co-injection of CM with unregulated opioids (colloquially termed "goofballing") may have important implications on risks of infections and overdoses (Al-Tayyib et al., 2017; Glick et al., 2020; Jones, Bekheet, et al., 2020; Meacham et al., 2016). A thorough understanding of trajectories, relationships, and shifts in substance use patterns over time can appropriately direct our research efforts, help us forecast trends in other settings and allow us to better predict potential harms, target interventions, and prevent important health consequences.

The city of Vancouver is situated on the West Coast of Canada and represents one of the largest urban settings for substance use in the country. Observations in our local context have previously presaged changes in patterns of unregulated drug use that have subsequently extended eastward (Canadian Centre on Substance Use and Addiction, 2020; Casey, 2016). The current study set out to describe shifting patterns of injection cocaine and CM use over time and relative to one another in Vancouver, Canada. We also set out to report descriptive data on both the price and availability of these substances over time.

2. Methods

The study obtained data from the Vancouver Injection Drug Users Study (VIDUS) and the AIDS Care Cohort to evaluate Exposure to Survival Services (ACCESS) studies, two open prospective cohort studies of people who use drugs in Vancouver, Canada. These cohorts have been described in detail in previous literature (Strathdee et al., 1997). Briefly, they are ongoing, community-recruited cohorts that have been running since 1996. VIDUS comprises HIV-negative people who have used injection drugs within the month prior to enrollment, while ACCESS consists of HIV-positive people who have used unregulated substances (other than or in addition to cannabis, which was illegal during the study period) either by injection or noninjection in the month prior to enrollment. Individuals in VIDUS who experience HIV seroconversion are transitioned to the ACCESS study. At the time of enrollment, all participants had to be ≥ 18 years old, live within the Greater Vancouver Regional District, and provide informed consent. Participants were recruited through street outreach, word of mouth, or self-referral. At baseline and semi-annually following their enrollment, they completed an interviewer-administered questionnaire that elicited information on demographics, drugs use patterns and behaviors, and other exposures. Blood samples were also drawn to establish HIV and hepatitis C virus serostatus and to assess disease progression among those with HIV. Study protocols were harmonized to allow for pooled analysis when appropriate. The study remunerated participants with \$40 per visit for their time. The cohort studies received annual ethics approval from the University of British Columbia/Providence Health Care research ethics board.

For this study, we used data gathered from June 2008 to November 2018. To focus on patterns in injection stimulant use, we restricted the sample to participants who had a history of injection drug use at baseline (either at time of entry into the cohorts, or at a new baseline defined by the initiation of injection drug use at any point during follow-up). Primary outcomes included injection powdered cocaine use (the predominant form of injected cocaine in Vancouver; yes vs. no) and injection CM use (yes vs. no) within the previous six months. The study considered a number of explanatory variables based on hypothesized or known associations with unregulated stimulant use, including month and calendar year of interview, median age, ethnicity (white vs. non-white), sex at birth (male vs. female), homelessness (yes vs. no), residence in the Downtown Eastside (an urban Vancouver neighborhood with a high concentration of substance use; yes vs. no), injection nonprescription opioid use (includes heroin and/or fentanyl; yes vs. no), non-injection crack use (yes vs. no), HIV positive (yes vs. no), and recent nonfatal

overdose (yes vs. no). All behavioural variables referred to activities within the previous six months unless otherwise specified.

The study obtained baseline sample characteristics at time of first entry into the cohort or at the beginning of our study date (June 2008), whichever was most recent. Next, we compared over time trends in self-reported injection cocaine use, injection CM use, and injection of any stimulant (i.e., either cocaine and/or CM) within the previous six months using the Cochrane-Armitage test for trend.

To obtain an understanding of the relationships between patterns of use and time, we started by using each individual interview month to identify the proportion of people who reported injection cocaine use per month. We repeated the same procedure for each of the explanatory variables (including injection CM use), in order to establish monthly proportions. We included age as the median age of individuals during each month and added month and year of the interview to control for the time period. The study then examined association of monthly proportions of injection cocaine use with other variables using bivariate and multivariable linear regression to identify factors associated with injection cocaine use on a per month basis, with parallel analyses done for injection CM use. We included only variables that were not indicated as collinear with other covariates, as assessed by variance inflation factor testing (VIF < 2.5), in the multivariable models.

Next, we used a time series analysis and cross-correlation approach to determine relationships between injection cocaine use and injection CM use over time. We tested for stationarity using the Augmented Dickey Fuller (ADF) and Kwiatkowski-Phillips-Schmidt-Shin (KPSS) tests. To satisfy the stationarity condition, we analyzed the change in proportions of substance use from previous month, rather than the total proportion of individuals who used the substances monthly (meaning that at each time t with proportion x_t of individuals who used the substance, we used $x_t - x_{t-1}$ as our explanatory variable). We assessed a cross-correlation function (CCF) to detect correlation between injection of CM and cocaine. The sample cross-correlation function (CCF) serves to identify the relationship between two time series $x(t)$ and $y(t)$. The approach aims to identify h associated with the largest correlation between $x(t+h)$ and $y(t)$: a negative h indicates x leads y , while a positive h indicates x lags y . For the purposes of this analysis x represents injection CM use and y represents injection cocaine use. We had no pre-specified hypotheses due to the exploratory nature of this analytic approach.

Finally, as a further descriptive analysis, we trended reports on the price of both cocaine and CM by those who reported use of these substances over the same time period, as well as responses to whether a participant stated they could obtain either substance within a short (i.e. < 90 min) time frame.

The study staff performed all statistical models using R version 3.6.3 (R Core Team, Vienna, Austria) and all p -values were two-sided.

3. Results

The study included a total of 2056 participants with a history of injection drug use during the study period, 2008–2018. The median duration of follow-up per participant was 6.4 years (Q1–Q3 = 2.3–9.6). The median age of participants was 42 years old (Q1–Q3 = 35–49) at baseline, with a mean age of 44 years old among all participants in 2008 increasing to 49 years old by 2018. Of those included 64 % were male ($n = 1312$), and 46 % ($n = 945$) were of white ethnicity. At baseline, 25 % ($n = 514$) of participants reported use of injection CM within the past six months, and 37 % ($n = 757$) reported use of injection cocaine within the past six months. Additionally, 52 % reported injection nonprescription opioid use, and 70 % reported noninjection crack use. See Table 1 for a complete description of baseline characteristics.

As Fig. 1 shows, between 2008 and 2018 per annum rates of reported injection cocaine use declined significantly from 45 % to 18 % ($Z = -16.3$, $p < 0.001$), while per annum rates of reported injection CM increased from 17 % to 32 % ($Z = 21.6$, $p < 0.001$). When we combined

Table 1
Baseline characteristics of participants ($n = 2056$).

Characteristic	Total (%)
Median age (IQR 1–3)	42.2 (34.6–48.6)
Sex at birth	
Male	1312 (63.8)
Female	743 (36.1)
White ethnicity	
Yes	945 (46.0)
No	1100 (53.5)
Homelessness ^a	
Yes	767 (37.3)
No	1281 (62.3)
Downtown Eastside residence ^a	
Yes	1410 (68.6)
No	646 (31.4)
Injection heroin use ^a	
Yes	1061 (51.6)
No	990 (48.2)
Injection cocaine use ^a	
Yes	757 (36.8)
No	1294 (62.9)
Injection meth use ^a	
Yes	514 (25.0)
No	1536 (74.7)
Non-injection crack use ^a	
Yes	1447 (70.4)
No	607 (29.5)
HIV	
Yes	817 (39.7)
No	1239 (60.3)
Overdose ^a	
Yes	166 (8.1)
No	1884 (91.6)

Numbers may not add to 100 % due to missing or incomplete data.

^a Denotes activities or situations referring to the past 6 months.

these variables into any reported injection stimulant use (either cocaine and/or CM), the overall prevalence did not change over time ($Z = 1.3$, $p = 0.21$).

Next, in a bivariate analysis comparing proportion of injection cocaine use with time on a per month basis, the proportion reporting monthly injection cocaine use was negatively associated with time ($\beta = -0.165$, 95 % CI = -0.187 , -0.142 , adjusted $R^2 = 0.63$; [Appendix 1](#)), whereas the proportion reporting monthly injection CM use was positively associated with time ($\beta = 0.187$, 95 % CI = 0.165 , 0.208 , adjusted $R^2 = 0.71$; [Appendix 2](#)). Multivariable analysis including all variables that were not considered multicollinear ([Appendix 3](#)) showed that

recent CM injection was negatively associated with recent cocaine injection on a per month basis ($\beta = -0.609$, 95 % CI = -0.750 , -0.467 ; [Appendix 1](#)).

Subsequently, a cross-correlation analysis of patterns of reported injection cocaine and injection CM use across the study sample found that increased injection CM use predicted decreased likelihood of injection cocaine use over time, with a peak lag of 12 months ($p = 0.002$; [Fig. 2](#)). We emphasized the largest absolute correlation value to determine the strongest correlation between these two time series.

Finally, in a descriptive analysis of self-reported changes in price over time using common unit of sale as the standardized measure, the average reported price per 0.5 g of cocaine increased by 176 % between 2008 and 2018 (from \$20.50 to \$36.00 CAD), while the price of 0.1 g of CM declined by 18 % over the same time period (from \$11.40 to \$9.30 CAD). The percentage of participants who stated that they could obtain cocaine within 90 min remained relatively stable at 93 % to 92 % over the course of the study, while the reported availability of CM within 90 min increased from 59 % to 90 % ([Appendix 4](#)).

4. Discussion

This study demonstrates opposing trends in the prevalence of reported injection stimulant use in Vancouver over the past decade, with rising rates of CM injection accompanied by a concomitant decline in rates of injection cocaine use. Furthermore, our results suggest that the overall prevalence of injection stimulant use within this cohort has remained relatively stable over time, but there has been an epidemiologic trend of moving from the use of injection cocaine toward injection CM.

Patterns of substance use in general are variable and may be impacted by combinations of individual factors (e.g., physical dependence and/or tolerance, medical/psychiatric co-morbidities, personal preferences, etc.) as well as socio-structural ones (e.g., social norms and peer groups, unregulated drug availability and cost, the social/economic/legal milieu, etc.) ([Rhodes, 2002](#)). While CM was traditionally a drug with high rates of geographic and cultural variability, its use is now rising in many populations of people who use unregulated drugs across North America ([Bach et al., 2020](#); [Ellis et al., 2018](#); [U.S. Department of Justice, 2019](#)). Though the reasons behind this are likely complex and multifactorial, the increasing availability of methamphetamine is notable, along with the associated decreasing price of the drug and an increasing purity that have been seen broadly over time elsewhere and are at least partly echoed in our results ([U.S. Department of Justice, 2019](#)). In our context, the extended half-life of methamphetamine

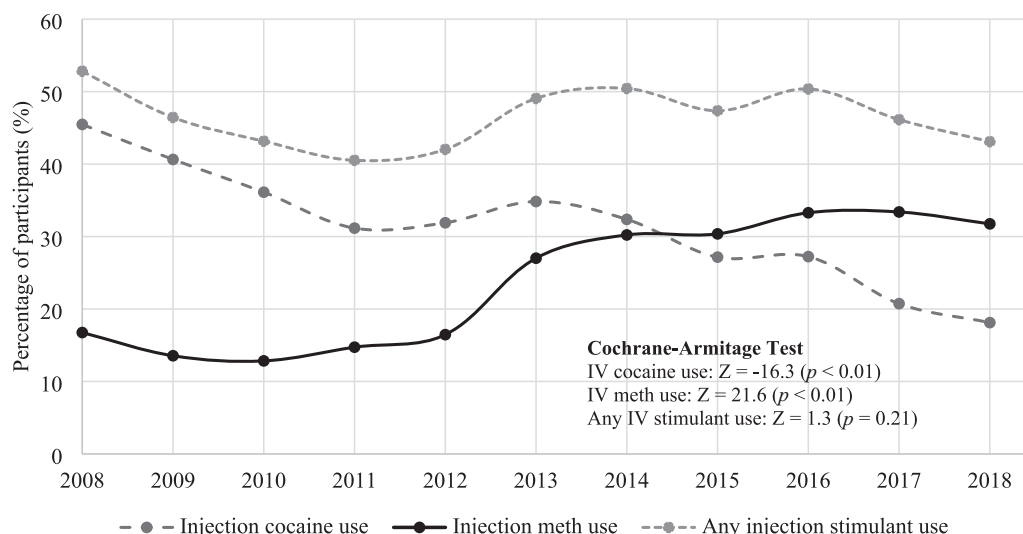


Fig. 1. Patterns of recent injection stimulant use over time ($n = 2056$).

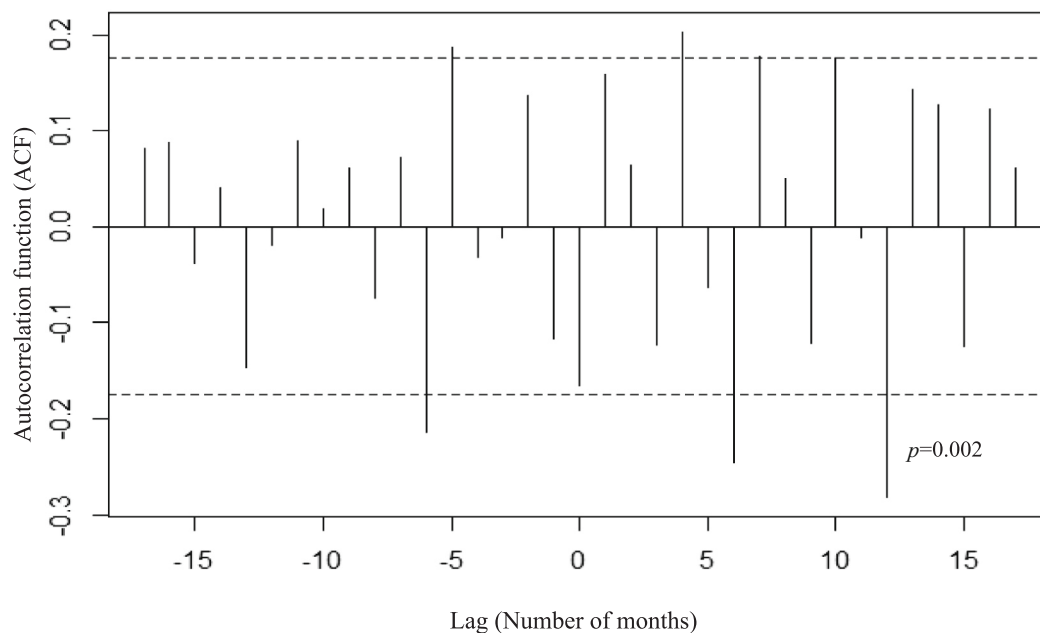


Fig. 2. Cross-correlation of monthly change in injection cocaine use and monthly change in injection methamphetamine use (2008–2018).

*Cross-correlation plot with vertical lines depicting positive or negative correlations between the changes in injection cocaine use and changes in injection methamphetamine use over monthly intervals. Lag refers to the time period in months between related changes in these activities. Hashed line signifies threshold for significance ($p < 0.05$), with the strongest detected relationship seen at a lag of 12 months ($p = 0.002$), indicating injection methamphetamine use predicted decreased likelihood of injection cocaine use over time, with a peak lag between these two of 12 months.

**Note that the autocorrelation function (ACF) is reported here due to a lagged X axis.

compared to cocaine (10–12 h, compared to 0.5–2 h for cocaine) may provide further motivation for a switch, particularly in the case of injection drug use and in environments where substance must often be concealed due to the harsh penalties imposed by criminalization (Ciccarone, 2011). CM use may also be especially prevalent among people experiencing homelessness; research has previously described it as a survival tool to stay awake, to avoid harm, and to cope with experiences such as cold and hunger (Fast et al., 2014).

Though the rise in injection CM use that we describe here is consistent with previous North American reports, this is one of the first reports of a significant decline in injection cocaine use in the same population, as overall trends in Canada and the United States suggest that cocaine use has been relatively steady or even increasing over time (though substance use trends remain highly geographic and heterogeneous, and declines in crack/cocaine injection alone have been noted in other specific populations) (Government of Canada, 2017; Substance Abuse and Mental Health Services Administration, 2020; U.S. Department of Justice, 2019; Roy et al., 2017). Within Vancouver, Canada, rates of injection cocaine use were exceptionally high in the 1990s, serving as a key driver in the severe HIV outbreak that afflicted the city at that time (Tyndall et al., 2003). Since then, however, its use has been on a steady decline, while rates of noninjection crack use have been rising over that same period (Urban Health Research Initiative, 2013). This transition toward a powerful, synthetic stimulant such as CM may potentially be analogous to the emergence of fentanyl, which has replaced heroin as the predominant form of opioid found in the unregulated drug supply in many North American settings (Volkow, 2021). In the case of fentanyl this shift has presumably been due to ease of production of a fully synthetic opioid molecule (rather than the semi-synthetic heroin, which requires natural production of an opium precursor), as well as ease of transportation of a highly potent, concentrated form of drug—a comparison that also reflects many of the similarities and differences between cocaine and CM (Karila et al., 2019). Of note, while the price of cocaine may have increased over time in our study setting, its availability was consistent throughout the study period.

These data suggest that lack of cocaine availability alone was not the primary driver of the observed increase in CM use, though the price change may have had some bearing on our observed results. The availability of robust drug purity measurements is limited in this setting (and not available longitudinally), but the most recent data from this drug market suggest that the unregulated stimulant supply remains relatively pure, with a median cocaine concentration in 2019 in the range of 85 %, and a median methamphetamine concentration of approximately 96 % (Long et al., 2020).

Consequences clearly attributable to this rise in CM use have not yet been reported in Vancouver, but its detection in overdose deaths has increased from 14 % in 2012 to 39 % in 2019 (British Columbia Coroners Service, 2021). In the context of an ongoing overdose crisis driven largely by fentanyl and other synthetic opioid analogues, we do not know how much, if any, of this rise may be related to the CM use itself, but, particularly given reports of increasing rates of methamphetamine and heroin/fentanyl co-injection, it warrants closer investigation (Al-Tayyib et al., 2017; Glick et al., 2020; Meacham et al., 2016). Our findings of an overall high prevalence of stimulant use are also concerning given the context of what we know about effective treatments for these disorders. Evidence-based tools are limited, with the strongest evidence supporting contingency management programs as the most effective approach for increasing rates of abstinence and retaining individuals in treatment (De Crescenzo et al., 2018). No evidence exists to suggest that either cocaine use or methamphetamine use is more amenable to treatment through psychosocial approaches, and pharmacologic options for treatment remain lacking (Vocci & Montoya, 2009). One recent meta-analysis suggested that prescription psychostimulants may have a role to play in some situations, but their effect appeared mostly limited to participants with cocaine use disorder, which is unfortunate given the trends reported here (Tardelli et al., 2020).

This study has a number of limitations that must be taken into consideration. First, though we attempted to explore the relationship between cocaine and CM injection over time in our study, it is an observational study with sampling occurring every six months only;

therefore, the study could not establish causal relationships (see [Appendix 5](#) for hypothesized relationships and directionality). Second, cross-correlation remains an exploratory analytic approach, thus firm conclusions are limited. Third, we have relied on self-reporting of sensitive data by our study participants, and thus cannot rule out a recall and response bias. We should note, however, that previous research has shown self-reporting of similar variables to be valid in comparable populations ([Darke, 1998](#)). Fourth, we cannot rule out a potential role for the covariation principle biasing measurement of self-reported substance use, whereby shifts in one pattern of use (e.g. cocaine injection) may predispose participants to perceive shifts in another pattern of use (e.g. CM injection) that do not truly exist ([Kelley, 1973](#)). Fifth, this cohort did not comprise a randomly recruited sample and contained a majority of participants from the Downtown Eastside of Vancouver, thus the generalizability of our study results to other contexts may be limited and inferences can only be made at a cohort level. Sixth, the average age of participants in this cohort increased over the duration of this study, thus a cohort effect (i.e. variations in the characteristics of a cohort over time) may also be contributing to our results. Finally, we only have data available on the use of injection stimulants within the past six months, and more granular data on specific injecting frequencies and patterns over shorter time periods cannot be inferred.

5. Conclusions

In a prospective cohort of people who inject unregulated drugs, the prevalence of reported injection CM use has risen markedly over the past decade, accompanied by a decrease in injection cocaine use. Importantly, the study observed no significant change in overall reported injection stimulant use prevalence. These results describe an epidemiologic shift toward the preferential use of a synthetic stimulant, and one that does not appear to be explained by the accessibility of these drugs alone. Further qualitative research should seek to better understand the socio-structural factors that have led to these observed changes over time, and the field urgently needs to develop tools to treat and reduce harm among a growing population of people who inject CM.

CRediT authorship contribution statement

Paxton Bach: Conceptualization, Methodology, Writing – original draft. **Lianping Ti:** Conceptualization, Methodology, Writing – review & editing. **Kanna Hayashi:** Conceptualization, Supervision, Writing –

review & editing. **Zishan Cui:** Methodology, Writing – review & editing. **M.-J. Milloy:** Supervision, Writing – review & editing. **Nadia Fairbairn:** Conceptualization, Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors have no competing interests to declare. MJM declares no personal financial relationship with the cannabis industry.

Acknowledgements

The authors thank the study participants for their contribution to the research, as well as current and past researchers and staff. The study was supported by the US National Institute on Drug Abuse (NIDA; U01-DA038886, U01-DA0251525). PB is supported by funds from Michael Smith Health Research BC, the British Columbia Centre on Substance Use, and the St. Paul's Foundation. For the project described he was also supported by the NIDA funded Research in Addiction Medicine Scholars (RAMS) Program (R25DA033211). LT is supported by a Michael Smith Health Research BC Scholar Award. KH holds the St. Paul's Hospital Chair in Substance Use Research and is supported in part by NIDA (U01DA038886), a Michael Smith Health Research BC Scholar Award, and the St. Paul's Foundation. MJM is supported by NIDA (U01-DA02351525), a New Investigator award from the Canadian Institutes of Health Research, and a Scholar Award from Michael Smith Health Research BC. His institution has received an unstructured gift from NG Biomed, a private firm seeking a license to produce cannabis, to support him. He holds the Canopy Growth professorship in cannabis science at the University of British Columbia, a position funded by unstructured gifts from Canopy Growth, a licensed producer of cannabis, and the Government of British Columbia's Ministry of Mental Health and Addictions. NF is supported in part by the US National Institutes of Health (2R25DA037756), the UBC Philip Owen Professorship in Addiction Medicine, and a Scholar Award from Michael Smith Health Research BC/St. Paul's Foundation.

Abbreviations

ACCESS AIDS Care Cohort to evaluate Exposure to Survival Services
CM crystal methamphetamine
VIDUS Vancouver Injection Drug Users Study

Appendix 1. Bivariate and multivariable linear regression analysis of factors associated with monthly prevalence of reporting injection cocaine use ($n = 2056$)

Characteristic	Unadjusted	Adjusted
	Slope β (95 % CI)	Slope β (95 % CI)
Time (per month)	−0.165 (−0.187, −0.142)*	–
Median age	−1.901 (−2.405, −1.397)*	–
Male sex	0.624 (0.359, 0.888)*	0.456 (0.226, 0.686)*
White ethnicity	−0.444 (−0.757, −0.131)*	−0.291 (−0.555, −0.026)*
Homelessness ^a	0.587 (0.397, 0.777)*	–
Downtown Eastside residence ^a	0.395 (0.209, 0.581)*	–
Injection methamphetamine use ^a	−0.574 (−0.706, −0.442)*	−0.609 (−0.750, −0.467)*
Injection heroin use ^a	−0.053 (−0.322, 0.216)	0.503 (0.271, 0.734)*
Non-injection crack use ^a	0.414 (0.365, 0.463)*	–
HIV positive	−0.358 (−0.629, −0.086)*	0.159 (−0.088, 0.406)
Overdose ^a	−1.068 (−1.302, −0.834)*	–

Items that showed multicollinearity with crystal methamphetamine use were not included in the adjusted model.

^a Behavioural variables were categorized as the proportion of all responses on a per month basis.

* $p < 0.05$.

Appendix 2. Bivariate and multivariable linear regression analysis of factors associated with monthly prevalence of reporting injection crystal methamphetamine use ($n = 2056$)

Characteristic	Unadjusted	Adjusted
	Slope β (95 % CI)	Slope β (95 % CI)
Time (per month)	0.187 (0.165, 0.208)*	–
Median age	2.182 (1.663, 2.700)*	–
Male sex	–0.588 (–0.876, –0.301)*	0.084 (–0.162, 0.330)
White ethnicity	0.330 (–0.010, 0.669)	–0.033 (–0.305, 0.239)
Homelessness ^a	–0.468 (–0.684, –0.252)*	–
Downtown Eastside residence ^a	–0.371 (–0.572, –0.170)*	–
Injection cocaine use ^a	–0.653 (–0.803, –0.503)*	–0.619 (–0.763, –0.475)*
Injection heroin use ^a	0.691 (0.431, 0.950)*	0.741 (0.529, 0.953)*
Non-injection crack use ^a	–0.442 (–0.495, –0.390)*	–
HIV positive	0.304 (0.012, 0.596)*	0.288 (0.043, 0.533)*
Overdose ^a	1.351 (1.138, 1.564)*	–

Items that showed multicollinearity with crystal methamphetamine use were not included in the adjusted model.

^a Behavioural variables were categorized as the proportion of all responses on a per month basis.

* $p < 0.05$.

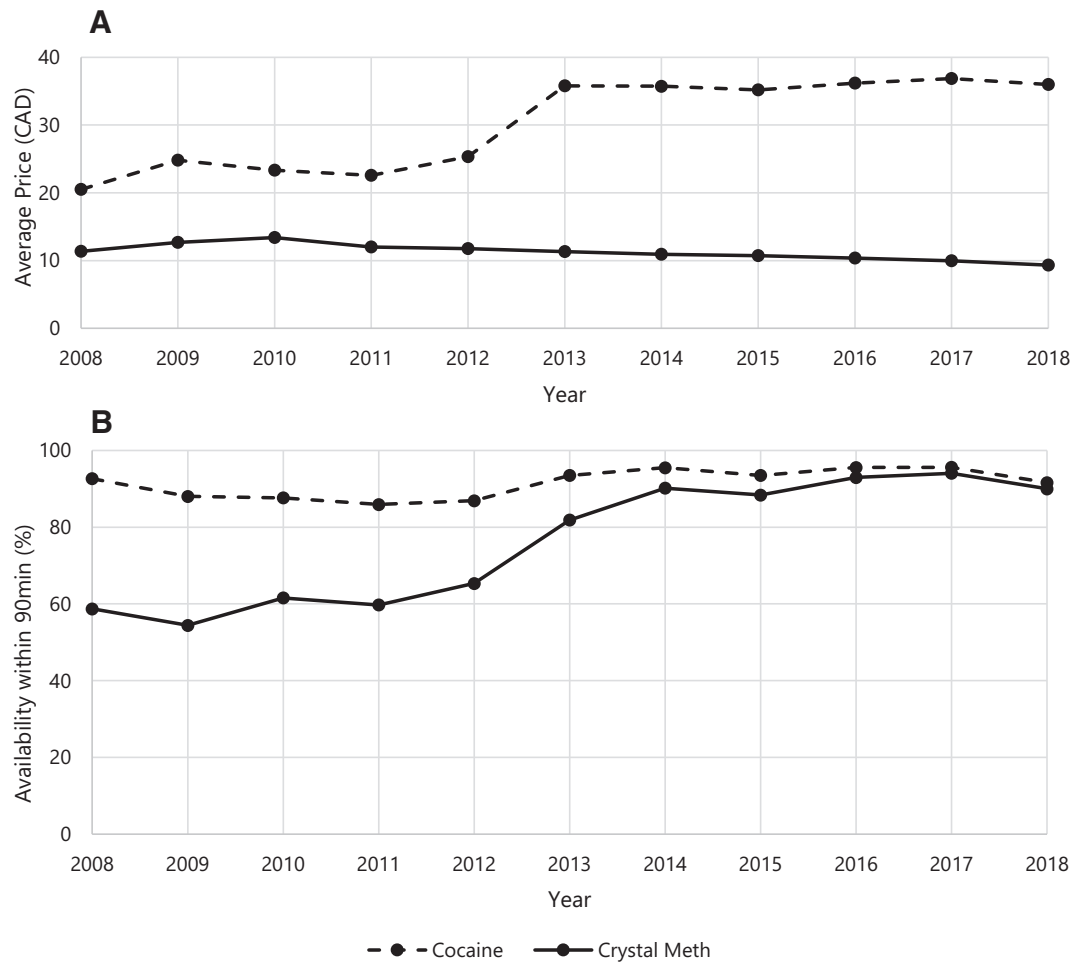
Appendix 3. Variance inflation factors to detect multicollinearity between predictors considered in a bivariable linear regression analysis of factors associated with reporting injection cocaine use, before and after restriction for variables collinear with methamphetamine use

Characteristic	VIF before restriction	VIF after restriction
Median age	7.23	–
Male sex	1.82	1.27 ^b
White ethnicity	1.52	1.33 ^b
Homelessness ^a	3.23	–
Downtown Eastside residence ^a	3.33	–
Injection methamphetamine use ^a	5.41	1.43 ^b
Injection heroin use ^a	2.11	1.47 ^b
Non-injection crack use ^a	16.95	–
HIV positive	2.01	1.56 ^b
Overdose ^a	3.63	–
Date (per month)	28.94	–

^a Behavioural variables were categorized on a per month basis.

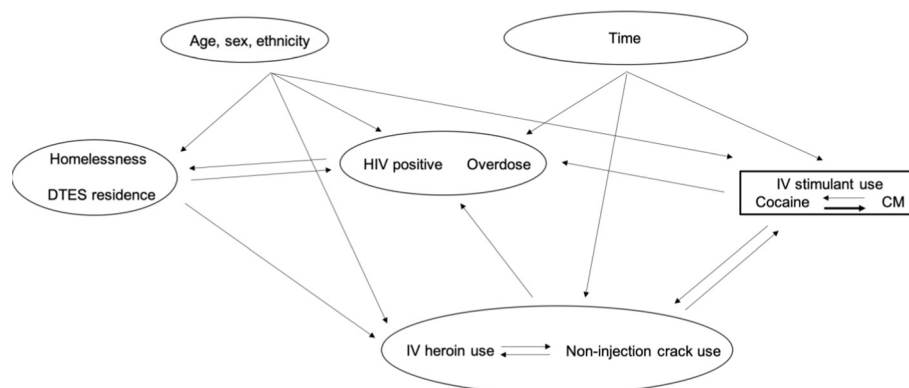
^b Included in the multivariable analysis.

Appendix 4. Average reported price and availability of cocaine and crystal methamphetamine over time (2008–2018)



(A) Average reported price per 0.5 g of cocaine and 0.1 g of crystal methamphetamine over time in Canadian dollars (CAD); (B) percentage of participants reporting availability within <90 min of cocaine and crystal methamphetamine over time.

Appendix 5. Directed acyclic graph describing hypothesized relationships between variables



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